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Applications Note

Genome analysis

Ktrim: an extra-fast and accurate adapter- and quality-trimmer for sequencing data

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Abstract

Motivation: Next-generation sequencing (NGS) data frequently suffer from poor-quality cycles and adapter contaminations therefore need to be preprocessed before downstream analyses. With the ever-growing throughput and read length of modern sequencers, the preprocessing step turns to be a bottleneck in data analysis due to unmet performance of current tools. Extra-fast and accurate adapter- and quality-trimming tools for sequencing data preprocessing are therefore still of urgent demand.

Results: Ktrim was developed in this work. Key features of Ktrim include: built-in support to adapters of common library preparation kits; supports user-supplied, customized adapter sequences; supports both paired-end and single-end data; supports parallelization to accelerate the analysis. Ktrim was ~2–18 times faster than current tools and also showed high accuracy when applied on the testing datasets. Ktrim could thus serve as a valuable and efficient tool for short-read NGS data preprocessing.

Availability and implementation: Source codes and scripts to reproduce the results descripted in this article are freely available at https://github.com/hellosunking/Ktrim/, distributed under the GPL v3 license.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

In next-generation sequencing (NGS) technology, the DNA molecules of interest are ligated with adapter sequences during the library preparation step in order to be recognized by the sequencers. However, when the DNA molecules are shorter than the read length of the sequencing run [which is common when analyzing plasma cell-free DNA (Sun *et al.*, 2018) or small RNA], the output reads will contain both the DNA molecules of interest and the adapter sequences. In addition, the quality of the sequencing reads usually drops in the tail cycles. Reads that are contaminated by sequencing adapters or poor-quality cycles frequently result in suboptimal downstream analyses; trimming of the adapters and poor-quality cycles of sequencing data thus become a common preprocessing procedure before downstream analyses. Various tools had been developed for adapter- and quality-trimming (Bolger *et al.*, 2014; Chen *et al.*, 2018; Jiang *et al.*, 2014; Martin, 2011; Sturm *et al.*, 2016); however, advances in modern sequencers have raised new challenges due to the ever-growing throughput and read length. As a result, the performance (e.g. speed and accuracy) of current tools becomes unsatisfactory and makes the preprocessing step a bottleneck in data analysis. Extra-fast and accurate adapter- and quality-trimming tools for NGS data preprocessing are thus still of demand.

In this work, a flexible NGS reads preprocessor, Ktrim, was developed. Ktrim supports both commonly used adapters in commercialized library preparation kits and user-supplied, customized sequences. Ktrim also supports both paired-end and single-end data, as well as parallelization for analysis acceleration. Ktrim was ~2–18 times faster than current tools and also showed high accuracy on the testing datasets. Ktrim could thus serve as a valuable and efficient preprocessor for short-read NGS data analysis.

2 Implementation

Ktrim provides both adapter- and quality-trimming of the sequencing data. The schematic workflow of Ktrim was shown in Supplementary Figure S1. In the current implementation, Ktrim performs quality-trimming first. For each read, Ktrim screens the quality of the sequencing cycles from its tail to head and stops at a position where the quality of the corresponding cycle is higher than the given threshold; then Ktrim will trim all the cycles after this one. If the remaining sequence is shorter than the minimum acceptable size (which parameter could be set by the users), this read will be discarded; otherwise, Ktrim will perform adapter-trimming on this read. Ktrim employs a dual-index scheme to search for adapters in the sequencing data. For paired-end reads, Ktrim uses the head 3 bp of the two adapters linked to read 1 and read 2, respectively, as indexes to find the potential ligating position ('seed's) of the adapters; for single-end data, however, Ktrim uses the head 3 bp and the very next 3 bp of the adapter

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Table 1. Performance comparison of Ktrim and current tools

	Ktrim	Trim galore	Trimmomatic	SeqPurge[a]
Paired-end 10 million reads				
Running time (single-thread)[b]	27.23 (0.22)	298.09 (28.18)	65.82 (2.65)	NA
Peak memory (single-thread)[c]	0.73	0.32	35.41	NA
Running time (4-threads)	11.29 (0.11)	89.96 (1.63)	20.72 (0.24)	207.30 (0.65)
Peak memory (4-threads)	2.63	1.21	35.96	0.44
Correct trimming (%)	96.09	81.85	87.76	>99.99
Over-trimming (%)[c]	3.82	17.51	0.00	<0.01
Miss-trimming (%)[d]	0.09	0.64	12.24	<0.01
Single-end 10 million reads				
Running time (single-thread)	18.52 (0.11)	124.55 (8.76)	33.36 (1.23)	NA
Peak memory (single-thread)	0.39	0.32	35.41	NA
Running time (4-threads)	7.26 (0.03)	31.04 (0.92)	9.43 (0.18)	NA
Peak memory (4-thread)	1.42	1.21	35.70	NA
Correct trimming (%)	93.81	81.65	86.12	NA
Over-trimming (%)	5.58	17.51	0.00	NA
Miss-trimming (%)	0.60	0.84	13.88	NA

[a]SeqPurge does not support single-thread mode and single-end data; [b]Wall-clock time shown as 'mean (SD)' in seconds based on 10 repeat runs; [c]Over-trimming refers to reads that do not contain adapters but get trimmed; [d]Miss-trimming refers to reads that contain adapters but are not trimmed.

linked to read 1 as the indexes. This dual-index approach could tolerate sequencing errors in the adapter sequences thus improves sensitivity. After finding the 'seed's, Ktrim searches for adapters from the left-most 'seed' cycle by comparing the sequencing data versus the adapters base-by-base until the end of the sequencing reads or the adapters, whichever occurs first. In this manner, Ktrim could handle both scenarios where adapters are completely included or partially covered in the sequencing data. If the number of mismatches identified during the comparison of sequencing reads and adapters is smaller than a dynamic threshold (which parameter could be set by the users), this 'seed' position will be called as the adapter ligation site, then Ktrim trims all the cycles after the 'seed' position and ignores all the rest 'seed's; otherwise, the next 'seed' position will be investigated until all the 'seed's get exhausted. After adapter-trimming, Ktrim checks the sizes of the reads again and only reports those longer than the minimum acceptable size specified by the user.

For performance evaluations, 10 million paired-end DNA reads were *in silico* generated. The read length was set to 100 bp while DNA insert size ranged from 10 to 200 bp. Hence, the dataset was consisted of reads that contain complete or partial adapters as well as those not contaminated by adapters. Meanwhile, random mutations were also introduced into the reads to simulate sequencing errors. Notably, the adapter information is known for these *in silico* reads, therefore allows accuracy evaluation and comparison of adapter-trimming algorithms. Furthermore, the read-1 also served as the testing dataset of single-end reads. Three widely used tools, Trim Galore [a wrapper of the Cutadapt program (Martin, 2011)], Trimmomatic (Bolger *et al.*, 2014) and SeqPurge (Sturm *et al.*, 2016), were collected for performance comparisons. In addition, a real dataset collected from the literature (Zhang *et al.*, 2016) was also analyzed. Detailed information (including codes) on simulation data generation and performance evaluations could be found in Supplementary Method.

3 Results

Ktrim is implemented in C++ and runs on GNU Linux/Unix platforms. Ktrim provides built-in support for commonly used adapters by Illumina and BGI sequencers while user-supplied, customized adapter sequences are also supported. Other major features of Ktrim includes supporting both paired-end and single-end sequencing data, as well as parallelization for analysis acceleration. Notably, Ktrim and SeqPurge also support multiple files as input while Trim Galore and Trimmomatic do not, which feature could be useful when the reads are generated from various lanes.

To evaluate the performance of Ktrim, both *in silico* generated and real datasets were employed. The analyses were carried out using a computing server equipped with Intel Xeon CPUs and standard 64-bit CentOS system. The results on *in silico* reads (incorporated with 1% errors) were shown in Table 1. Ktrim was ~2–18 times faster than the others. Moreover, Ktrim showed high accuracy, which was slightly lower than SeqPurge but much better than the other two. When error rate in simulation data increases, the accuracies of Ktrim and SeqPurge were marginally unaffected while remarkably decreased for Trim Galore and Trimmomatic (Supplementary Table S1). On the real dataset, Ktrim showed comparable speed advancement as on the *in silico* dataset. All the trimmers benefited in higher number of mappable reads than the raw data; moreover, Ktrim, Trim Galore and SeqPurge produced comparable results, which were higher than Trimmomatic (Supplementary Table S2).

In conclusion, this work presents Ktrim, a flexible, extra-fast and accurate adapter- and quality-trimming tool for sequencing data. The high performance of Ktrim may be attributed to its dual-index scheme, which not only tolerates sequencing errors in adapters for high sensitivity (i.e. Ktrim would only miss the adapters when both 'seed's contain errors, which probability was very low considering the error rates of current sequencers) but also limits the adapter searching to the 'seed's for speed-up. Ktrim could thus serve as a valuable and efficient preprocessor to accelerate short-read NGS data analysis.

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Conflict of Interest: none declared.

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